

# Practice Task 3. Advanced Excel Functions

## Description:

Practiced applying different Excel formulas such as VLOOKUP, mathematical, and statistical functions to analyze datasets, solve problems, and enhance the accuracy and reliability of calculations.

### IF

**What does it do?**  
It checks whether a condition is true or not and on the basis of that returns a value

**Syntax**  
`=if(Logical_Test, [value_if_true], [value_if_false])`

**Logical\_Test**  
This will be the condition which you want to check, so greater than, less than or equal to signs can be used for numbers. When you want to evaluate a condition based on **text value**, then use double quotes (") with equal to signs (Row 27 for example)

**Value\_if\_True**  
If the logical test holds, then excel will return this value

**Value\_if\_False**  
If the logical test does not hold, then excel will return this value

**Example**  
Using IF formula to identify if a student has passed an exam or not

| Student ID | Marks | Pass or Fail | Formula                                    |
|------------|-------|--------------|--------------------------------------------|
| S001       | 90    | Pass         | <code>=IF(B22&gt;=40,"Pass","Fail")</code> |
| S002       | 70    | Pass         | <code>=IF(B23&gt;=40,"Pass","Fail")</code> |

Logical - IF   Logical - AND   Logical - Nested IFs   Logical - OR   Logical - IFERROR   Text - Exact   Text - TEXT   Text - CONCATENATE   Text - TRIM   Text - SUBSTITUTE